

Sharp-cube scalar-budget obstruction and the Class-II gauge-null pivot

TECT verification-first repository · claim A12-CLASSII-SOURCE-SQUARE-REDUCTION

first issued 2026-07-21 · this version issued 2026-07-21 · v1.0

1. Purpose and scope

Exact negative result. For the strict dyadic sharp rectangular-cube filtration used by A10–A12,

$$M_6 \geq 8, \quad Q_6 \geq 8\sqrt{3}, \quad H_6 := M_6^4 Q_6^2 \geq 786432.$$

The A12 source-only target is $H_6 < 29.62571266025876$, so no certified upper enclosure of the separated constant can close that budget. The same witness proves that the coefficient-blind six-linear scalar envelope has norm at least 786432.

Surviving route. This does not refute the actual A11 source. The exact production matrix satisfies $B(X)\mathcal{J}X = 0$, while the A12 scalar envelope discarded this gauge-null identity, the output shell projection, and the determinant resolvent. The successor must retain those structures before any Fourier estimate.

Tier boundary. A12 remains scoped T4. No positive sextic reserve, stabilised log-Laplace estimate, Nelson bound, interacting measure, or tier promotion is claimed.

The setting is the fixed torus $\mathbb{T}_L^3$ with $L = 16$, $\alpha_L = 2\pi/L$, dyadic integers $N_j = 2^j$, and

$$S_j = \mathbf{1}_{\{\max_\alpha |n_\alpha| \leq N_j\}}.$$

The target space is the six-real representation of three complex fields and the derivative square function contains all three spatial directions. A complex scalar test is admissible by embedding its real and imaginary parts in two of the six real components.

2 The separated constants and the production target

A12 defined

$$M_6 = \sup_{f \neq 0} \frac{\|\sup_j |S_j f|\|_6}{\|f\|_6}$$

and

$$Q_6 = \sup_{f \neq 0} \frac{\left\| \left(\sum_j \frac{|DS_{j-1}f|^2}{1 + \alpha_L^2 (N_{j-1} + 1)^2} \right)^{1/2} \right\|_6}{\|f\|_6}.$$

Its valid qualitative reduction is

$$\sum_j \|G_j^* B(S_{j-1}\phi) DS_{j-1}\phi\|_2^2 \leq \frac{\beta_{\text{op}}^2}{c_{\text{sym}}} M_R^2 H_6 \|\phi\|_6^6. \quad (2.1)$$

At $p = 1.1$, $M_R = 1$, and production $\gamma = 1.62$, source-only absorption would require

$$H_6 < 29.62571266025876. \quad (2.2)$$

Equation (2.2) is a proposed threshold, not an established estimate.

3 Dyadic boundary modulation

Let $P_{\leq 0}$ be the one-dimensional Fourier projection onto frequencies $n \leq 0$. Hollenbeck and Verbitsky proved the exact Riesz projection norm

$$\|P_{\leq 0}\|_{L^6(\mathbb{T}) \rightarrow L^6(\mathbb{T})} = \csc(\pi/6) = 2. \quad (3.1)$$

Choose a finite trigonometric polynomial g and let $F(x) = g(x_1)g(x_2)g(x_3)$. For a dyadic N larger than the Fourier degree of g , set

$$U_N F(x) = e^{i\alpha_L N(x_1+x_2+x_3)} F(x).$$

For every relative frequency in the support of F ,

$$-N \leq N + n_a \leq N \iff n_a \leq 0.$$

Consequently

$$S_N U_N F = U_N (P_{\leq 0}^{\otimes 3} F). \quad (3.2)$$

The input fits below the next terminal cube $2N$, so (3.2) is an admissible finite-cutoff witness. Finite trigonometric polynomials are dense in L^6 ; using product near-extremisers for (3.1) gives

$$M_6 \geq \|P_{\leq 0}^{\otimes 3}\|_{6 \rightarrow 6} \geq 2^3 = 8. \quad (3.3)$$

Use the single square-function term whose previous cutoff is N . If $h = P_{\leq 0}^{\otimes 3} F$, then

$$D(U_N h) = U_N \{i\alpha_L N(1, 1, 1)h + Dh\}.$$

After division by $\sqrt{1 + \alpha_L^2(N+1)^2}$ and passage to large dyadic N , the norm ratio tends

$$\sqrt{3}, \frac{\|h\|_6}{\|F\|_6}.$$

Taking the same product near-extremisers yields

$$Q_6 \geq 8\sqrt{3}. \quad (3.4)$$

Combining (3.3)–(3.4),

$$\boxed{H_6 \geq 8^4 (8\sqrt{3})^2 = 3 \cdot 8^6 = 786432.} \quad (3.5)$$

This is more than 2.65×10^4 times the target (2.2).

4 The coefficient-blind six-linear route also fails

Define the scalar-envelope constant

$$\mathfrak{P}_6 := \sup_{f \neq 0, J} \frac{\sum_{j \leq J} \frac{1}{1 + \kappa_j^2} \int_{\mathbb{T}_L^3} |S_{j-1} f|^4 |DS_{j-1} f|^2}{\|f\|_6^6}. \quad (4.1)$$

The single boundary term of the preceding witness converges to

$$3 \frac{\|P_{\leq 0}^{\otimes 3} F\|_6^6}{\|F\|_6^6}.$$

Therefore

$$\boxed{\mathfrak{P}_6 \geq 3 \cdot 8^6 = 786432.} \quad (4.2)$$

Even a one-axis witness gives $\mathfrak{P}_6 \geq 2^6 = 64$, already above the production target. Thus replacing $M_6^4 Q_6^2$ by a direct six-linear theorem after the pointwise estimate

$$|B(u)Du| \leq \beta_{\text{op}}|u|^2|Du|$$

cannot repair the budget. The pointwise multiplier bound in A12 controls symbol size and qualitative Marcinkiewicz variation; it does not state that the multilinear L^6 operator norm is one.

5 Independent finite rational countercertificate

The manifest contains a Gaussian-integer polynomial supported on $[-10, 10]$. Its coefficients share denominator 10^{10} , so Parseval gives the sixth power of the L^6 norm exactly: if $c * c * c$ is the triple discrete convolution, then

$$\left\| \sum_n c_n z^n \right\|_6^6 = \sum_k |(c * c * c)_k|^2. \quad (5.1)$$

For the stored polynomial, the exact ratio obeys

$$\frac{\|P_{\leq 0}g\|_6^6}{\|g\|_6^6} = 3.9475110593975523284 \dots \quad (5.2)$$

Tensorisation and the carrier derivative give the elementary certified lower bound

$$3 \left(\frac{\|P_{\leq 0}g\|_6^6}{\|g\|_6^6} \right)^3 = 184.5403419180373503 \dots > 29.62571266025876. \quad (5.3)$$

This finite exact countercertificate fires the gate without relying on a numerical approximation to the sharp norm (3.1). Equation (3.5) is the stronger asymptotic theorem.

6 Why the actual Class-II source is not refuted

Let $X \in \mathbb{R}^6$ be the realification of the complex field and let

$$\mathcal{J} = \begin{pmatrix} 0 & -I_3 \\ I_3 & 0 \end{pmatrix}$$

be the global phase generator. For each embedded Pauli generator, write

$$p_A = 2S_A X, \quad v_A = 2(S_A - q_A I)X.$$

The real symmetric matrix S_A commutes with $\mathcal{J}$. Hence $S_A \mathcal{J}$ is skew-symmetric and

$$p_A^T \mathcal{J} X = 0, \quad v_A^T \mathcal{J} X = 0. \quad (6.1)$$

Since

$$B(X) = \sum_A \{a, p_A p_A^T + b(p_A v_A^T + v_A p_A^T) + c, v_A v_A^T\},$$

equation (6.1) gives the exact fixed-floor identity

$$\boxed{B(X) \mathcal{J} X = 0.} \quad (6.2)$$

The large derivative in the boundary-modulation witness is precisely the common phase direction $\mathcal{J}X$. The actual matrix removes it, whereas the coefficient-blind envelope (4.1) retains it.

Two further structures were also discarded in passing to (2.1): the actual source contains the output shell projection of $B(S_{j-1}\phi)DS_{j-1}\phi$, and the noncentral determinant contains a positive resolvent. A viable successor must retain at least (6.2) and the output shell localisation, and should test whether the resolvent gives an additional gain.

For completeness, positivity of the production Class-II quadratic form is explicit rather than inferred from $b > 0$:

$$\begin{aligned} a &= 0.0044999999999998875 > 0, & c &= 0.002343749999999414 > 0, \\ ac - b^2 &= 7.031249999996483 \times 10^{-6} > 0. \end{aligned} \tag{6.3}$$

7 Consequence and proof order

The gate **A12-CLASSII-SHARP-CUBE-L6-VECTOR-NORM-ENCLOSURE** is closed negatively: an upper enclosure exists abstractly, but every valid one is at least (3.5), so it cannot satisfy (2.2). The coefficient-blind six-linear replacement is eliminated by (4.2).

The next gate is

A12-CLASSII-COEFFICIENT-AWARE-SHELL-LOCALISED-SOURCE-BOUND.

It must retain the exact B , its gauge-null direction, the output shell, all three relative-phase directions, the fixed floor, and the production determinant convention. Only after that bound has an explicit production constant may the full sextic budget and the A11 stabilised log-Laplace gate be revisited. Changing to a smooth filtration is a separate fallback because it would require rederiving the dependent determinant construction.

8 Numerical verification

Run

```
python codes/foundations/
  a12_classii_sharp_cube_budget_obstruction_verify.py
```

The primary audit derives the production threshold, sharp lower bounds, coefficient positivity, and gauge-null identity. The non-importing audit uses the finite Gaussian-integer countercertificate and reconstructs the current identities directly. The integrated verifier checks hashes and cross-route equality. FFT sampling is not used as proof of an infinite-scale upper bound.

9 Devil’s-advocate review

1. **DISMISSED – the modulation is not dyadic.** The carrier is chosen as a power of two. Equation (3.2) is exact on every fixed finite polynomial once the dyadic carrier exceeds its degree.
2. **DISMISSED – the witness falls outside the terminal cutoff.** Its largest frequency is $N+m < 2N$ for $N > m$, so the next dyadic terminal cube contains the full input while the previous cube supplies the boundary term.
3. **DISMISSED – complex tests are inadmissible for a six-real field.** A complex scalar is embedded as two real components. Fourier projection, Euclidean norm, and derivatives agree under this embedding.

